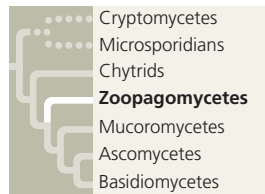


Zoopagomycetes



Most of the 900 species of **zoopagomycetes**, fungi in the phylum Zoopagomycota, live as parasites or as commensal (neutral) symbionts of animals; some are parasites of other fungi or protists.

Zoopagomycetes form filamentous hyphae and reproduce asexually by producing nonflagellated spores. Some zoopagomycetes induce insects that they parasitize to perch near the top of plants; the insects subsequently die and fungal spores are released to infect new victims (**Figure 31.14**). Sexual reproduction, where known, involves the formation of a durable structure called a zygosporangium, which houses and protects the zygote.

The loss of flagellated spores in the zoopagomycetes and other fungal lineages may have been associated with a



Figure 31.14
A fly covered with fungal hyphae.
This fly was killed by the zoopagomycete *Entomophthora muscae*, also known as the “fly death fungus,” for obvious reasons.

transition to life on land. Basal fungal lineages had flagellated spores, enabling dispersal through water. In contrast, zoopagomycetes and all of their closest fungal relatives (the clade consisting of the mucoromycetes, ascomycetes, and

Figure 31.15 The life cycle of the mucoromycete *Rhizopus stolonifer* (black bread mold).

